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A Profit-Maximizing Model Marketplace with Differentially Private Federated Learning

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Abstract

Existing machine learning (ML) model marketplaces generally require data owners to share their raw data, leading to serious privacy concerns. Federated learning (FL) can partially alleviate this issue by enabling model training without raw data exchange. However, data owners are still susceptible to privacy leakage from gradient exposure in FL, which discourages their participation. In this work, we advocate a novel differentially private FL (DPFL)-based ML model marketplace. We focus on the broker-centric design. Specifically, the broker first incentivizes data owners to participate in model training via DPFL by offering privacy protection as per their privacy budgets and explicitly accounting for their privacy costs. Then, it conducts optimal model versioning and pricing to sell the obtained model versions

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still challenging due to its non-convexity and integer constraints. We hence propose efficient algorithms, and their performances are both theoretically guaranteed and empirically validated.

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